



Koneru Lakshmaiah Education Foundation

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DEPARTMENT OF CSE

PROJECT ABSTRACT SUBMISSION

COURSE: PROJECT EVALUATION (24CS2235F)

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TITLE OF THE PROJECT	Smart Home Rule Compiler	
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Abstract

The **Smart Home Rule Compiler** is a domain-specific compiler designed to translate simple, human-readable smart home automation rules into executable device instructions. The system allows users to define rules based on conditions such as temperature, motion, time, and device status. These rules are processed through key compiler phases including lexical analysis, parsing, syntax tree construction, intermediate code generation, optimization, and code generation. The system also identifies syntax errors, redundant rules, and conflicting conditions. A virtual smart home environment demonstrates the execution of the compiled instructions. The project combines compiler design concepts with smart home automation to provide a practical and interactive application.

Signature of the Guide

HOD